

Andi Zhang

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</> Summary

I am currently an Assistant Professor at the University of Warwick. I obtained my PhD from the University of Cambridge, where I proposed the **Probabilistic Adversarial Attack**, a principled probabilistic perspective to adversarial attacks. After my PhD, I developed the **Concept-based Adversarial Attack**, which leverages the probabilistic perspective of adversarial attacks to extend traditional single-image adversarial attacks to broader semantic concepts or entities, expanding the applicability of adversarial methods.

🔗 Work Experience

University of Warwick Coventry, United Kingdom
Assistant Professor Mar. 2026 -

- Concentrating on AI safety and LLM safety.

University of Manchester Manchester, United Kingdom
Postdoc Research Associate in Machine Learning | Supervisor: Prof. Samuel Kaski Nov. 2024 - Feb. 2026

- Supported by UKRI Generative AI Hub.
- Proposed concept-based adversarial attack.

🏛️ Education

Trinity College, University of Cambridge
PhD in Computer Science Oct. 2019 - Dec. 2024

- Thesis Title: **Anomalous Inputs in Deep Learning: a Probabilistic Perspective**
- Supervisor: Dr. Damon Wischik
- Keywords: Probabilistic Generative Models, Large Language Models, Out-of-distribution Detection, Adversarial Attack

MASt in Mathematics (Part III of the Mathematical Tripos) Oct. 2018 - Jun. 2019
• Relevant modules: Statistical Learning in Practise, Convex Optimization, Astrostatistics, Bayesian Modelling and Computation, Modern Statistical Methods

MPhil in Advanced Computer Science Oct. 2017 - Jun. 2018
• Relevant modules: Machine Learning for Natural Language Processing, Algebraic Path, Deep Learning for Natural Language Processing, Advanced Functional Programming, Probabilistic Machine Learning

School of Computer Science, University of Manchester
BSc (Hons) in Computer Science and Mathematics Sep. 2014 - Jun. 2017
• Relevant modules: Machine Learning, Quantum Computing, Compilers, Convex Optimization, Probability, Statistical Methods, Functional Analysis, Topology, Number Theory

🔗 Internships

MediaTek Research Cambridge, United Kingdom
Deep Learning Intern | Host: Dr. Alberto Bernacchia Aug. 2024 - Nov. 2024

- Participated to a project in accelerating the sampling of diffusion models by proposing a new second-order estimation of the posterior.

University College London London, United Kingdom
Visitor | Hosts: Dr. David Barber and Dr. Mingtian Zhang July. 2023 - Oct. 2023

- Worked on a new image generative model that based on Diffusion models.
- Worked on Amortized LoRA, a more principle and probabilistic way to generate LoRA parameters.

Kirin AI Group, Huawei Cambridge, United Kingdom
Research Intern (Part-time) | Host: Mr. Arnau Raventos Nov. 2021 - Oct. 2022

- Conducted a comprehensive evaluation of various traditional and AI-based frame interpolation methods, providing insights for the product department to leverage in their decision-making processes.

- Contributed to an AI global illumination project, successfully implemented various samplers within the Mitsuba 3 framework.

Noah's Ark Lab, Huawei

Shenzhen / Beijing, China

Research Intern | Hosts: Dr. Yitong Sun and Dr. Shifeng Zhang

Sep. 2020 - Sep. 2021

- Co-developed the local autoregressive model, which significantly advanced the state-of-the-art in Out-Of-Distribution (OOD) detection and lossless compression. The empirical evidence supporting this enhancement was recognized and published in NeurIPS 2021.
- Contributed to a computer vision project through researching deep learning uncertainty for active learning applications. The outcomes of this work were recognized and published in BMVC 2021.
- Contributed to an indoor positioning project leveraging WiFi signal strength. I specifically conducted research on Deep Gaussian Processes, leading our team to enhance indoor positioning accuracy. The successful results of this work were published in IPIN 2021.

University of Cambridge

Cambridge, United Kingdom

Course Supervisor (Teaching Assistant)

Oct. 2019 - Dec. 2024

- Supervised IA Algorithms (2020 Lent, 2023 Lent), IA Machine Learning in Real World (2020 Lent), IA Introduction to Probability (2020 Easter, 2021 Easter), IB Further Java (2020 Michaelmas), IB Data Science (2022 Michaelmas, 2023 Michaelmas), IB Further Graphics (2022 Michaelmas) and II Machine Learning and Bayesian Inference (2023 Lent)

MRC Biostatistics Unit, University of Cambridge

Cambridge, United Kingdom

Research Assistant | Hosts: Dr. Sofia Villar and Dr. David Robertson

Jul. 2019 - Sep. 2019

- Developed a simulation system to tackle the multi-armed bandit problem (MABP) in clinical trial settings and pioneered a unique test statistic for binary response MABP situations. Although the time constraints prevented this work from being formalized into a publication, it provided substantial practical and theoretical insight.

Tencent (WeChat AI)

Beijing, China

R & D Intern | Host: Dr. Jie Zhou

Jul. 2018 - Sep. 2018

- Engaged in comprehensive exploration and research in deep reinforcement learning and evolution strategies. While this research did not culminate in a tangible output, the experience gained in these cutting-edge fields has proven invaluable.

Amazon (Alexa)

Cambridge, United Kingdom

Applied Scientist Intern | Host: Dr. Emilio Monti

Jun. 2017 - Sep. 2017

- Employed the Seq2Seq model, the predecessor of the Transformer, with the Attention mechanism for semantic parsing tasks. Semantic parsing involves translating natural language into a structured internal language used for database or knowledge base querying.

Robotics Institute, Carnegie Mellon University

Pittsburgh, United States

Student Intern | Host: Dr. Stelian Coros

Jun. 2016 - Sep. 2016

- Contributed to the implementation of a general stance controller for 3D-printable robotic creatures. The controller's adaptability enables it to be used with any human-designed multi-foot robots, hence its designation as 'general'.

Publications (Last / Corresponding Author)

Hierarchical Orthogonal Residual Spread for Precise Massive Editing in LLMs

ICASSP 2026 Oral

Xiaojie Gu, Guangxu Chen, Yuheng Yang, Jingxin Han, **Andi Zhang**

In the 2026 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)

FedMC: Federated Manifold Calibration

ICLR 2026

Yanbiao Ma, Wei Dai, Gaoyang Jiang, Wanyi Chen, Chenyue Zhou, Yiwei Zhang, Fei Luo, Junhao Wang, **Andi Zhang**

In the 14th International Conference on Learning Representations (ICLR)

Compositional Attribute Imbalance in Vision Datasets

AAAI 2026

Yanbiao Ma, Jiayi Chen, Wei Dai, Dong Zhao, Zeyu Zhang, Yuting Yang, Bowei Liu, Jiaxuan Zhao, **Andi Zhang**

In the 40th Annual AAAI Conference on Artificial Intelligence (AAAI)

Publications (First / Co-first Author)

Concept-based Adversarial Attack: a Probabilistic Perspective

ICLR 2026

Andi Zhang, Xuan Ding, Steven McDonagh, Samuel Kaski

In the 14th International Conference on Learning Representations (ICLR)

Your Finetuned Large Language Model is Already a Powerful OOD Detector

AISTATS 2025

Andi Zhang, Tim Z. Xiao, Weiyang Liu, Robert Bamler, Damon Wischik

In Proceedings of the 28th International Conference on Artificial Intelligence and Statistics

Constructing Semantics-Aware Adversarial Examples with a Probabilistic Perspective

NeurIPS 2024

Andi Zhang, Mingtian Zhang, Damon Wischik

In the 38th Annual Conference on Neural Information Processing Systems, 2024

SR-ODD: Out-of-Distribution Detection via Sample Repairing

Arxiv Preprint

Rui Sun*, **Andi Zhang***, Haiming Zhang*, Yao Zhu, Ruimao Zhang, Zhen Li

<https://arxiv.org/abs/2305.18228>

Falsehoods that ML researchers believe about OOD detection

NeurIPS 2022 Workshop

Andi Zhang, Damon Wischik

In ML Safety Workshop, 36th Conference on Neural Information Processing Systems

Out-of-Distribution Detection with Class Ratio Estimation

NeurIPS 2022 Workshop

Mingtian Zhang*, **Andi Zhang***, Tim Z. Xiao, Yitong Sun, Steven McDonagh

In ML Safety Workshop, 36th Conference on Neural Information Processing Systems

On the Out-of-distribution Generalization of Probabilistic Image Modelling

NeurIPS 2021

Mingtian Zhang*, **Andi Zhang***, Steven McDonagh

In Neural Information Processing Systems annual meeting, 2021

Other Publications

Improved Denoising Diffusion Probabilistic Models with Efficient Non-Diagonal Covariance Modeling

TMLR

Rui Xia*, Ayan Das*, Artem Artemev, **Andi Zhang**, Guillaume Hennequin, Alberto Bernacchia

In Transactions on Machine Learning Research (TMLR)

Noise Diffusion for Enhancing Semantic Faithfulness in Text-to-Image Synthesis

CVPR 2025

Boming Miao, Chunxiao Li, Xiaoxiao Wang, **Andi Zhang**, Rui Sun, ZiZhe Wang, Yao Zhu

In the IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR)

Improving Probabilistic Diffusion Models With Optimal Diagonal Covariance Matching

ICLR 2025 Oral

Zijing Ou*, Mingtian Zhang*, **Andi Zhang**, Tim Z. Xiao, Yingzhen Li, David Barber

In the 13th International Conference on Learning Representations (ICLR)

Measuring Uncertainty in Signal Fingerprinting with Gaussian Processes Going Deep

IPIN 2021

Ran Guan, **Andi Zhang**, Mengchao Li, Yongliang Wang

In International Conference on Indoor Positioning and Indoor Navigation, 2021

Towards dynamic and scalable active learning with neural architecture adaption for object detection

BMVC 2021

Fuhui Tang, Dafeng Wei, Chenhan Jiang, Hang Xu, **Andi Zhang**, Wei Zhang, Hongtao Lu, Chunjing Xu

In British Machine Vision Conference, 2021

Solipsistic Reinforcement Learning

ICLR 2021 Workshop

Mingtian Zhang*, Peter Noel Hayes*, Tim Z. Xiao, **Andi Zhang**, David Barber

In Self-Supervision for Reinforcement Learning Workshop - International Conference on Learning Representations, 2021

Academic Service

I have served as a reviewer for the following conferences:

- NeurIPS (2024, 2025, 2026)
- ICML (2025, 2026)
- ICLR (2025, 2026)
- AISTATS (2023, 2024, 2025, 2026)
- CVPR (2025, 2026)
- ICCV / ECCV (2025, 2026)
- ACM MM (2025)
- AAAI (2026)
- COLM (2026)

*Equal contribution.